

Project Report - Software Installation Request Automation in ServiceNow

1. INTRODUCTION

Project Overview

This project automates software installation requests using ServiceNow.

Purpose

To reduce manual work by automating request submission, approvals, fulfillment and notifications.

2. IDEATION PHASE

Problem Statement

Employees request software through emails causing delays.

Empathy Map Canvas

Focuses on employee needs, faster approvals and transparency.

Brainstorming

Centralized service catalog, automated approvals, workflow and notifications.

3. REQUIREMENT ANALYSIS

Customer Journey Map, Solution Requirements, Data Flow Diagram and Technology Stack (ServiceNow, Flow Designer, Service Catalog, Business Rules).

4. PROJECT DESIGN

Problem Solution Fit

Automates the complete software installation lifecycle.

Proposed Solution

Service Catalog + Flow Designer + Approval + Fulfillment.

Solution Architecture

Employee → Service Catalog → Approval → IT Support → Completion.

5. PROJECT PLANNING & SCHEDULING

Sprint planning, backlog, user stories and implementation schedule.

6. FUNCTIONAL & PERFORMANCE TESTING

Verified request submission, approval, workflow execution, task creation and completion.

7. RESULTS

Output screenshots are shown below.

8. ADVANTAGES & DISADVANTAGES

Advantages: Automation, transparency, tracking.

Disadvantages: Depends on ServiceNow and approvals.

9. CONCLUSION

The project successfully automates software installation requests.

10. FUTURE SCOPE

Multi-level approvals, asset management, AI recommendations.

11. APPENDIX

GitHub Link

Demo Link

Source Code

7.1 Output Screenshots

Application (Licensed Software Installation Management)

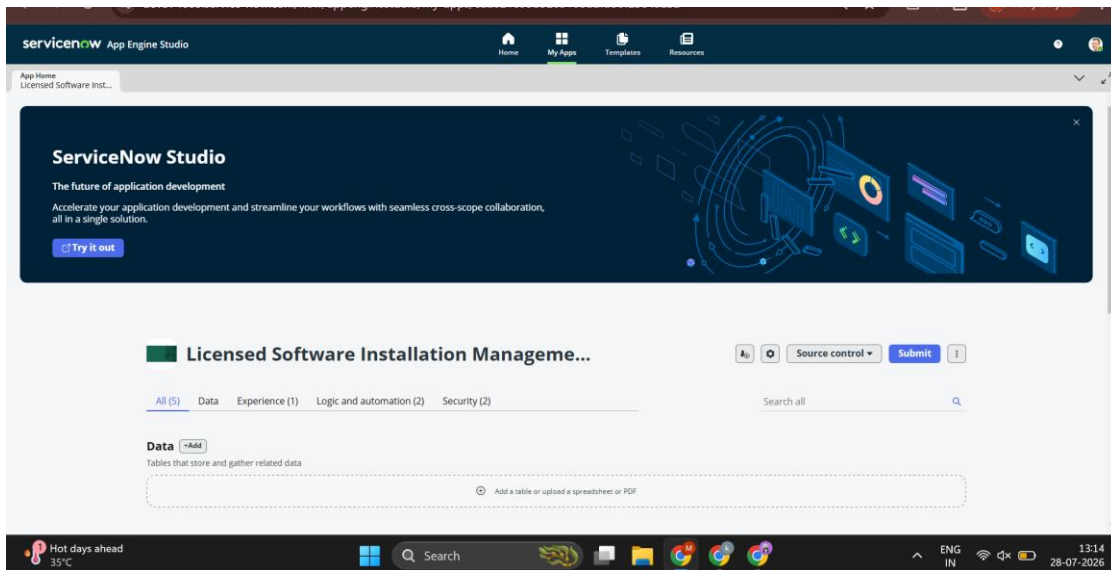


Figure: Application (Licensed Software Installation Management). This screenshot shows the application (licensed software installation management) screen of the Software Installation Request Automation project.

Catalog Item form

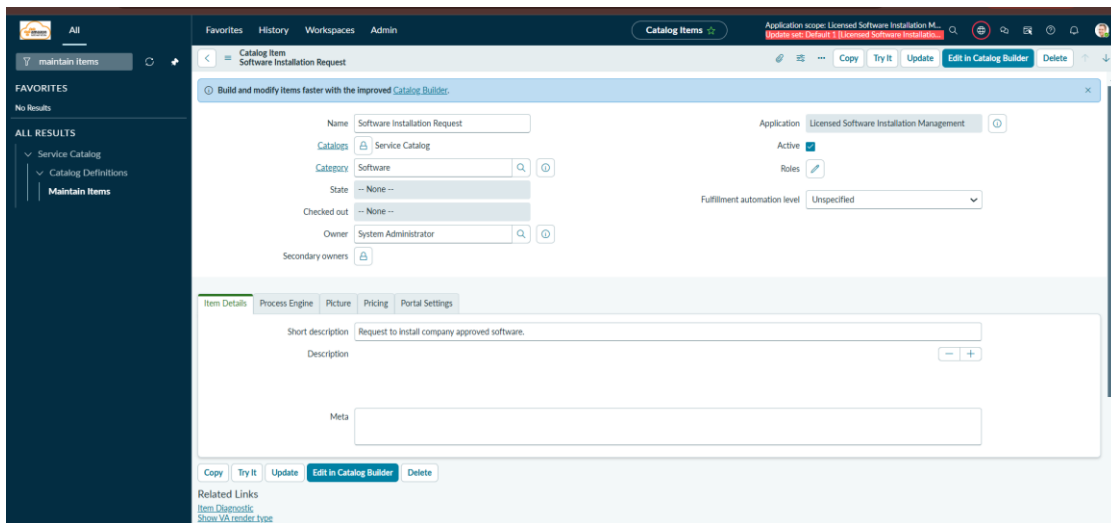


Figure: Catalog Item form. This screenshot shows the catalog item form screen of the Software Installation Request Automation project.

REQ order 2

The screenshot shows the 'My Request' page for request RITM0010014. The title is 'Request to install company approved software.' The request is in the 'Open' state, created just now, and last updated just now. The item is 'Software Installation Request' requested for 'System Administrator' in a quantity of 1. The stage is 'Assess or Scope Task'. The 'Activity' tab is selected, showing a comment from 'System Administrator' at 'just now' stating 'RITM0010014 Created'. The top navigation bar includes links for Knowledge, Catalog, Requests, System Status, Cart, Tours, and the user profile 'System Administrator'.

Figure: REQ order 2. This screenshot shows the req order 2 screen of the Software Installation Request Automation project.

Req items form

The screenshot shows the 'Requested Item' form for RITM0010013. The form includes fields for Number (RITM0010013), Item (Software Installation Request), Request (REQ0010013), Requested for (System Administrator), Due date (2026-07-30 10:21:54), and Configuration item. On the right, there are fields for Opened (2026-07-28 10:21:54), Opened by (System Administrator), Stage (Assess or Scope Task), State (Pending), Quantity (1), Estimated delivery, Backordered (checkbox), and Order Guide. Below these is a 'Variables' section with three required questions: 'What software do you need?' (Microsoft), 'Specify version (if required):' (2022), and 'Why do you need this software?' (Required for project development). There is also a 'Select urgency level' dropdown set to 'Normal'. At the bottom, there is a 'Work notes' section. The top navigation bar includes 'Follow', 'Update', and 'Delete' buttons. The bottom status bar shows the temperature (35°C), weather (Mostly cloudy), and the date/time (13:30, 28-07-2026).

Figure: Req items form. This screenshot shows the req items form screen of the Software Installation Request Automation project.

SP Form filling

The screenshot shows the 'Software Installation Request' form in ServiceNow. The form is titled 'Software Installation Request' and includes a description: 'Request to install company approved software.' and 'Allows employees to request software installation. Fulfills via Software Support Team.' The form contains several sections: 'Indicates required' with a red asterisk, 'What software do you need?' with a text field containing 'Microsoft', 'Specify version (if required):' with a text field containing '2022', 'Why do you need this software?' with a text field containing 'For complete the project development', 'Select urgency level:' with a dropdown menu showing 'Normal', and 'Add attachments' with a file upload button. On the right side, there is a sidebar with 'Quantity: 1', 'Delivery Time: 2 Days', and buttons for 'Add to Cart', 'Save as Draft', and 'Order Now'.

Figure: SP Form filling. This screenshot shows the sp form filling screen of the Software Installation Request Automation project.

Variables tab

The screenshot shows the 'Variables' tab for the 'Software Installation Request' catalog item. The tab is titled 'Variables (4)' and includes a search bar and a table of variables. The table has columns for 'Type', 'Question', and 'Order'. The variables are listed as follows:

Type	Question	Order
Single Line Text	What software do you need?	100
Single Line Text	Specify version (if required).	200
Multi Line Text	Why do you need this software?	300
Select Box	Select urgency level.	400

At the bottom of the table, there is a pagination bar showing '1 to 4 of 4'.

Figure: Variables tab. This screenshot shows the variables tab screen of the Software Installation Request Automation project.

approver req

Requested Item
RTM0010013

Work notes

Comments (Customer visible) ☐ Post

Activities: 2

System Administrator
State: Pending was Open
Field changes: 2026-07-28 10:29:37

System Administrator
Impact: 3 - Low
Opened by: System Administrator
State: Open
Field changes: 2026-07-28 10:21:54

Update Delete

Catalog Tasks (2) Approvers (1)

Approval for = RTM0010013

State	Approver	Comments	Created
Requested	Dev Manager		2026-07-28 13:30:10

1 to 1 of 1

No templates are available. Create A New One?

Figure: approver req. This screenshot shows the approver req screen of the Software Installation Request Automation project.

attaching the workflow to catalog items

Variable - Select urgency level.

Question Annotation Type Specifications Default Value Auto-populate Permission Availability

Specify the Question that explains the options available to the end user when ordering the Item

* Question: Select urgency level.

* Name: select_urgency_level

Conversational label

Tooltip: Choose how urgently you need software

Copy Update Delete

Related Links

Run Point Scan

SN Units Versions (3)

Question Choices Order Search

Question = Select urgency level.

Text	Value	Order	Inactive
Normal	normal	100	false
High	high	200	false
Critical	critical	300	false

1 to 3 of 3

No templates are available. Create A New One?

Figure: attaching the workflow to catalog items. This screenshot shows the attaching the workflow to catalog items screen of the Software Installation Request Automation project.

business rule script

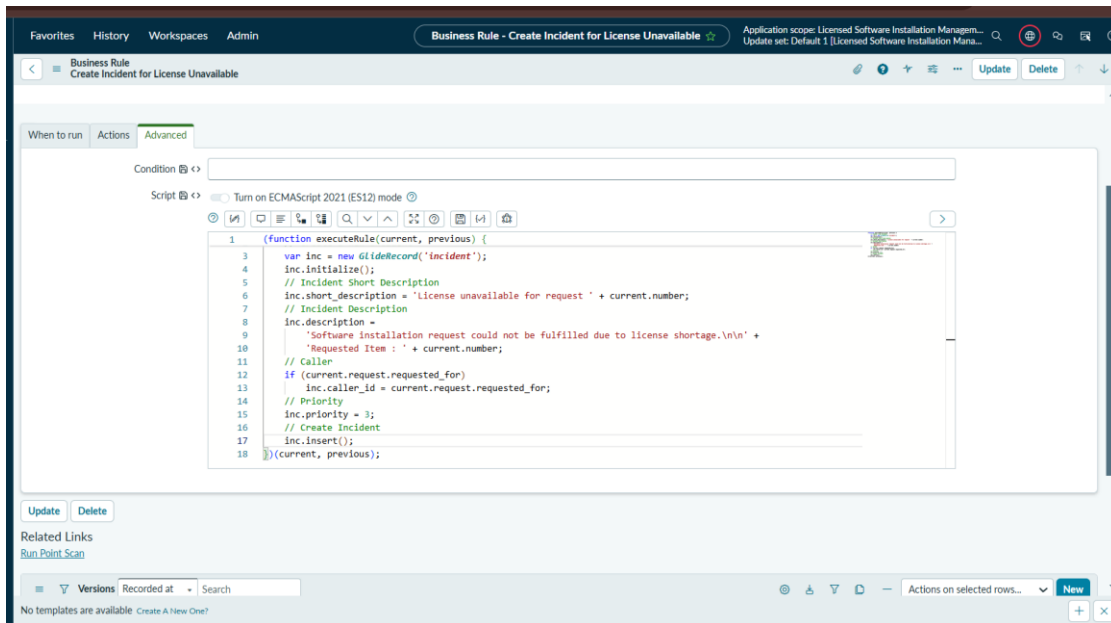


Figure: business rule script. This screenshot shows the business rule script screen of the Software Installation Request Automation project.

business rules

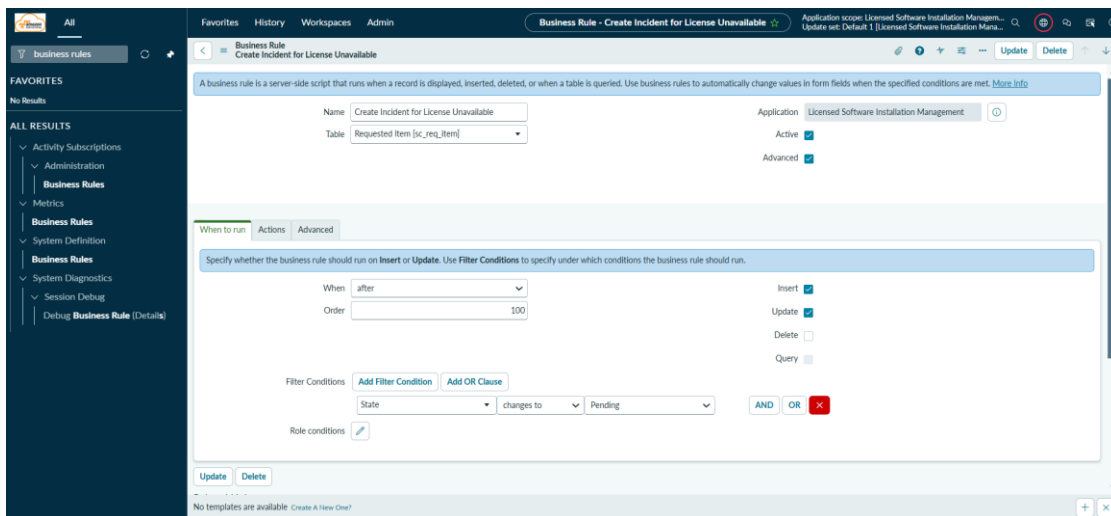


Figure: business rules. This screenshot shows the business rules screen of the Software Installation Request Automation project.

catalog clinet script(2)

Favorites History Workspaces Admin

Catalog Item - Software Installation Request

Application scope: Licensed Software Installation M...
Update set: Default 1 Licensed Software Installation...

Copy Try It Update Edit in Catalog Builder Delete

Press Alt+Q for help

Add relevant tags to the Meta field using comma-separated list of tags. These tags will be used while searching the item. Applicable only for keyword search when AI Search is enabled.

Meta

Copy Try It Update Edit in Catalog Builder Delete

Related Links

Item Diagnostic
Show VA render type
Run Point Scan
SN Utils Versions (26)

Variables (4) Variable Sets Catalog UI Policies (1) Catalog Client Scripts (2) Available For Not Available For Categories (1) Catalogs (1) Catalog Data Lookup Definitions Related Articles Related Catalog Items

Assigned Topics

for text Search

Actions on selected rows... New

Catalog Item - Software Installation Request

Name	Active	Table	View	Type	Updated
Urgency Change Notification	true			onChange	2026-07-25 20:34:39
Validate License Justification	true			onSubmit	2026-07-25 20:36:55

1 to 2 of 2

No templates are available Create A New One?

Figure: catalog clinet script(2). This screenshot shows the catalog clinet script(2) screen of the Software Installation Request Automation project.

incident form for req

Favorites History Workspaces Admin

Incident - INC0010046

Application scope: Licensed Software Installation Manag...
Update set: Default 1 Licensed Software Installation Mana...

Discuss Follow Update KITS Resolve Delete Responses

View: Self Service

Number INC0010046

* Caller System Administrator

Watch list

* Short description License unavailable for request RITM0010013

Related Search Results

Comments (Customer visible)

Post

Activities: 1

System Administrator

Impact 3 - Low
Incident state New
Opened by System Administrator
Priority 4 - Low

Field changes • 2026-07-28 10:29:38

Update KITS Resolve Delete Responses

Related Links

KITS
Show SLA Timeline
Repair SLAs

Templates: Incident Call Type Major Incident

Figure: incident form for req. This screenshot shows the incident form for req screen of the Software Installation Request Automation project.

req item approved

The screenshot displays the 'Requested Item' form in ServiceNow. The form is titled 'Requested Item' with the ID 'RITM0010014'. The 'View' is set to 'Self Service'. The 'Number' field contains 'RITM0010014' and the 'Due date' is '2026-07-30 13:40:19'. The 'Approval' status is 'Approved' and the 'Quantity' is '1'. The 'Stage' is 'Assess or Scope Task'. Below this, a table shows the 'Delivery Plan Status for: Request to install company approved software.' with columns for 'Task', 'Duration', 'Expected Start', 'Started', and 'Completed'. The table contains two rows: 'Assess or Scope Task' (1 Day, 2026-07-28, 2026-07-28) and 'Provide requested service' (1 Day, 2026-07-29). The 'Description' field contains the text: 'Allows employees to request software installation. Fulfills via Software Support Team.' The form also includes a 'Description' field with a rich text editor and a 'No templates are available. Create a New One?' message at the bottom.

Task	Duration	Expected Start	Started	Completed
Assess or Scope Task	1 Day	2026-07-28	2026-07-28	
Provide requested service	1 Day	2026-07-29		

Figure: req item approved. This screenshot shows the req item approved screen of the Software Installation Request Automation project.

service portal req home page

The screenshot shows the 'Software Installation Request' form in the ServiceNow service portal. The form is titled 'Software Installation Request' and has a subtitle 'Request to install company approved software.' The description is 'Allows employees to request software installation. Fulfills via Software Support Team.' The form includes several required fields: 'What software do you need?' (with an example: 'Microsoft Office, Visual Studio Code, IntelliJ IDEA'), 'Specify version (if required)' (with an example: '2022, 17.0, 1.20'), 'Why do you need this software?' (with an example: 'Required for final year project development.'), and 'Select urgency level' (with a dropdown menu set to 'Normal'). There is also an 'Add attachments' section with a 'Choose a file' button and a note 'Copy and paste clipboard files here.' On the right side, there is a 'Quantity' field set to '1', a 'Delivery Time: 2 Days' indicator, and buttons for 'Add to Cart', 'Save as Draft', and 'Order Now'. A 'Required information' section is also visible with a red warning message 'What software do you need?'.

Figure: service portal req home page. This screenshot shows the service portal req home page screen of the Software Installation Request Automation project.

status oending

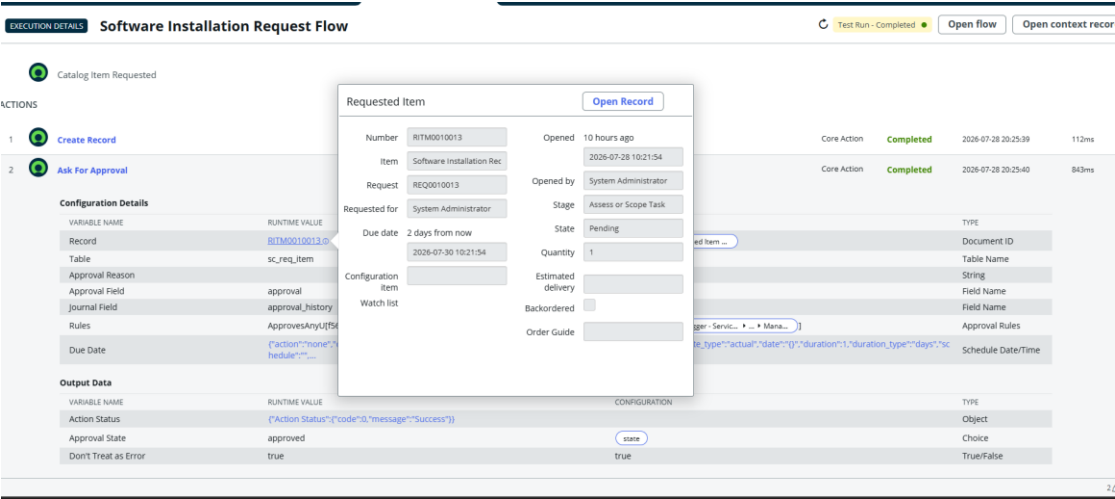


Figure: status oending. This screenshot shows the status oending screen of the Software Installation Request Automation project.

summary of req order

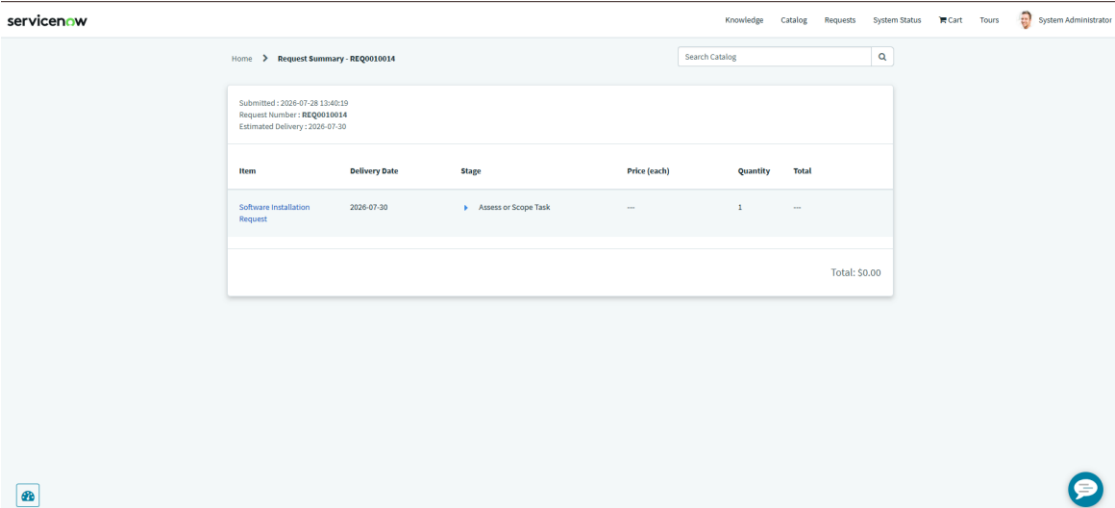


Figure: summary of req order. This screenshot shows the summary of req order screen of the Software Installation Request Automation project.

workflow 1

Workflow Studio | Software Installation Request Flow | Active

TRIGGER

Service Catalog

Trigger: Service Catalog

Advanced Options

Delete Cancel Done

ACTIONS Select multiple

- 1 Create Catalog Task Record
- 2 Ask For Approval on Requested Item

+ Add an Action, Flow Logic, or Subflow

ERROR HANDLER If an error occurs in your flow, the actions you add here will run.

Data Collapse All

- Flow Variables
- Trigger - Service Catalog
 - Requested Item Record Record
 - Run Start Time UTC DateTime
 - Table Name Table Name
 - Run Start Date/Time DateTime
- 1 - Create Record
 - Catalog Task Record Record
 - Catalog Task Table Table
 - Action Status Object
- 2 - Ask For Approval
 - Approval State Choice
 - Action Status Object

Figure: workflow 1. This screenshot shows the workflow 1 screen of the Software Installation Request Automation project.

workflow 2

Workflow Studio | Software Installation Request Flow | Active

ACTIONS Select multiple

- 1 Create Catalog Task Record

Action Properties

Action: Create Record

Action Inputs

* Table Catalog Task [sc_task]

* Fields

Parent Trigger - Requested Item

Short description Install requested software

Assignment group IT Support

Add field value

Delete Cancel Done

Data Collapse All

- Flow Variables
- Trigger - Service Catalog
 - Requested Item Record Record
 - Run Start Time UTC DateTime
 - Table Name Table Name
 - Run Start Date/Time DateTime
- 1 - Create Record
 - Catalog Task Record Record
 - Catalog Task Table Table
 - Action Status Object
- 2 - Ask For Approval
 - Approval State Choice
 - Action Status Object

Figure: workflow 2. This screenshot shows the workflow 2 screen of the Software Installation Request Automation project.

workflow 3

Software Installation Request Flow Active

Action Properties

Action: Ask For Approval

Action Inputs

Record: Trigger -> Requested item

Table: Requested item [sc_req_item]

Approval Reason: Approval

Approval Field: Approval history

Journal Field: Approval history

Rules: Approve When: Trigger - Service... -> Manual...

Due Date: None

Data

- Flow Variables
 - Trigger - Service Catalog
 - Requested item Record
 - Run Start Time UTC Date/Time
 - Table Name Table Name
 - Run Start Date/Time Date/Time
 - 1 - Create Record
 - Catalog Task Record Record
 - Catalog Task Table Table
 - 2 - Ask For Approval
 - Approval State Choice
 - Action Status Object

Figure: workflow 3. This screenshot shows the workflow 3 screen of the Software Installation Request Automation project.

workflow after manager approval

Software Installation Request Flow Test Run - Waiting Cancel flow Open flow Open context record

How Action Details

FLOW STATISTICS

Run as: System Administrator Open flow logs

TRIGGER

Catalog item Requested

ACTIONS

Step	Action	State	Start time	Duration
1	Create Record	Completed	2026-07-28 13:30:09	1009ms
2	Ask For Approval	Waiting	2026-07-28 13:30:10	2072ms

ERROR HANDLER

Figure: workflow after manager approval. This screenshot shows the workflow after manager approval screen of the Software Installation Request Automation project.

workflow for after the manager approval

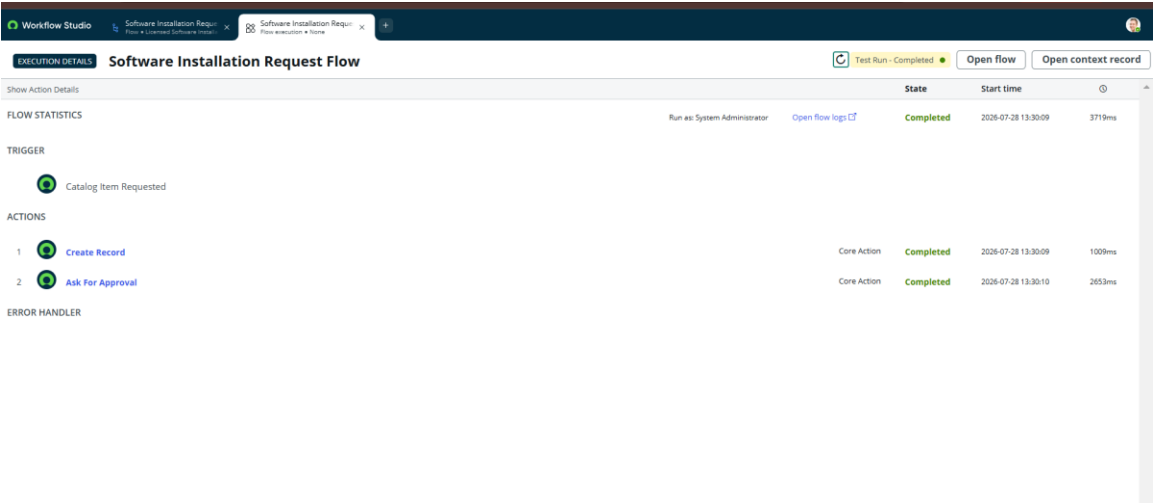


Figure: workflow for before the manager approval. This screenshot shows the workflow for before the manager approval screen of the Software Installation Request Automation project.

final output

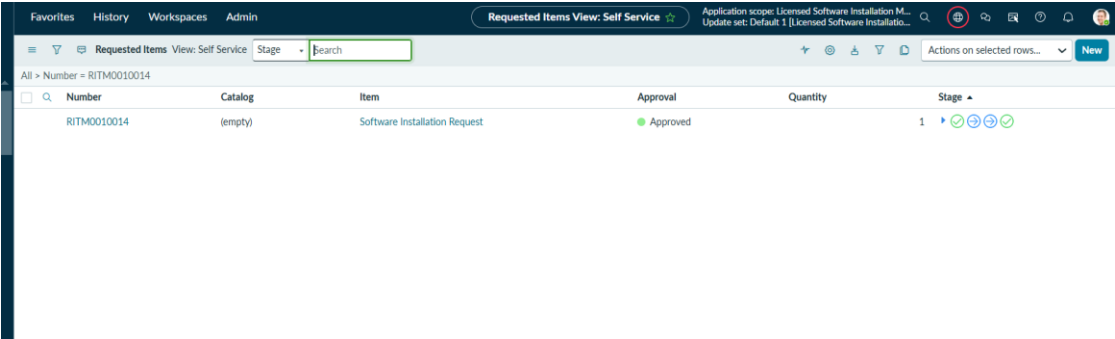


Figure: final output. This screenshot shows the final output screen of the Software Installation Request Automation project.